

Program : Diploma in Information Technology	
Course Code : 4263	Course Title: Advanced Web Development
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To develop web applications of PHP and MySQL.

Course Prerequisites:

Topic / Description	Course Code	Course Title	Semester
Knowledge of HTML, CSS and JavaScript		Fundamentals of Web Development	3
Basic Knowledge of MySQL		Database Management System Lab	3

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concept of PHP	11	Understanding
CO2	Develop web applications using PHP and MySQL	16	Applying
CO3	Illustrate AJAX and jQuery	17	Applying
CO4	Illustrate Bootstrap	14	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concept of PHP.		
M1.01	Define Server-Side Scripting, Define PHP, Summarize evolution of PHP	2	Remembering
M1.02	Summarize Environment creation for various platforms	2	Understanding
M1.03	Summarize Operators in PHP	1	Understanding
M1.04	Summarize Conditional Statements in PHP	1	Understanding
M1.05	Summarize Loops in PHP	2	Understanding
M1.06	Summarize Arrays in PHP	2	Understanding
M1.07	Summarize Functions in PHP.	1	Understanding
Contents: Define Server-Side Scripting-, Define PHP- Evolution of PHP-Discuss PHP Environment creation for various platforms-Syntax of PHP-Variables and Data Types-Operators in PHP-Conditional Statements in PHP- Loops in PHP-Arrays in PHP-Functions in PHP			
CO2	Develop web applications using PHP and MySQL		
M2.01	Summarize PHP global variables	1	Understanding
M2.02	Summarize HTML FORM handling using PHP	1	Understanding
M2.03	Summarize Time management in PHP	1	Understanding
M2.04	Define Session in PHP, Summarize Session in PHP (creation, modification and destroy)	1	Applying
M2.05	Define Cookies in PHP, Summarize Cookies in PHP (creation, modification and destroy)	1	Applying
M2.06	Summarize connection between PHP and MySQL	2	Applying

M2.07	Summarize Perform Insertion operations on MySQL table	2	Applying
M2.08	Summarize Perform Update operations on data in MySQL tables	2	Applying
M2.09	Summarize Fetch and Display data from the tables.	2	Applying
M2.10	Create applications to perform various database operations	3	Applying
	Series Test – I	1	

Contents: PHP Global Variables (Superglobals)- FORM handling using PHP-Time management in PHP-Session and Cookies in PHP.
Discuss the connection between PHP and MySQL Database- Perform Insertion operations on MySQL table- Perform Update operations on data in MySQL tables-Fetch and Display data from the tables.

CO3	Illustrate AJAX and jQuery.		
M3.01	Define AJAX and its needs, List Application areas of AJAX	1	Remembering
M3.02	Summarize the difference between HTTP and XMLHTTP	1	Remembering
M3.03	Summarize the creation of XMLHttpRequest object in JavaScript, Summarize the Generation of requests with XMLHttpRequest object	2	Applying
M3.04	Summarize readystatechange event in XMLHttpRequest object, summarize how to Send requests to server in synchronous & asynchronous methods	2	Understanding
M3.05	Summarize Get response from server; Summarize how to Manage text response and XML response.	2	Understanding
M3.06	Define jQuery, List need for jQuery, List advantages of jQuery	1	Remembering
M3.07	Show how to integrate jQuery into a webpage	1	Understanding
M3.08	Summarize jQuery syntax, Define ready function, jQuery selectors	2	Understanding
M3.09	Summarize jQuery effects (hide, slide, fade, toggle, stop)	3	Applying
M3.10	Summarize jQuery animations	2	Applying

AJAX: -Define AJAX-Needs of AJAX-Application areas of AJAX, Differentiate HTTP and XMLHttpRequest-Creat XMLHttpRequest object in JavaScript-Generate requests with XMLHttpRequest object-Describe readystatechange event in XMLHttpRequest object-Send requests to server in synchronous & asynchronous methods- Get response from server- Managing text response and XML response.

jQuery: -Define jQuery-List need for jQuery-List advantages of jQuery-Show how to integrate jQuery into a webpage - jQuery syntax, define ready function- jQuery effects: -

hide, slide, fade, toggle, stop- jQuery animations.			
CO4	Illustrate Bootstrap.		
M4.01	Define Bootstrap, List Advantages of Bootstrap, Download Bootstrap	2	Remembering
M4.02	Summarize Container & Container fluid and jumbotron in Bootstrap 3	2	Remembering
M4.03	Summarize buttons and Icons in Bootstrap 3	1	Understanding
M4.04	Summarize Multi column Layout in Bootstrap 3	1	Understanding
M4.05	Summarize Navigation in Bootstrap 3	1	Understanding
M4.06	Summarize Carousel, modal, Tooltip and Popover in Bootstrap 3	2	Understanding
M4.07	Summarize Forms, Form Inputs and Input sizing Bootstrap 3	2	Understanding
M4.08	Summarize working with images in Bootstrap 3	1	Understanding
M4.09	Summarize Tables in Bootstrap 3	2	Applying
	Series Test – II	1	
Contents: Bootstrap: -Define Bootstrap, List Advantages of Bootstrap, Download Bootstrap. Container & Container fluid – jumbotron -buttons - Icons - Multi column Layout- Navigation- Forms and Modal Bootstrap 3 working with images and tables in Bootstrap 3			

Text / Reference

T/R	Book Title/Author
T1	PHP and MySQL web development -Fifth edition-Luke Welling, Laura Thomson
T2	Step by step Bootstrap 3-A Quick guide responsive web development using Bootstrap 3- Riwanto Megosinarso

Online Resources

S.I.No	Website Link
1	https://www.w3schools.com/php/
2	https://www.tutorialspoint.com/php/index.htm
3	https://www.w3schools.com/bootstrap/default.asp